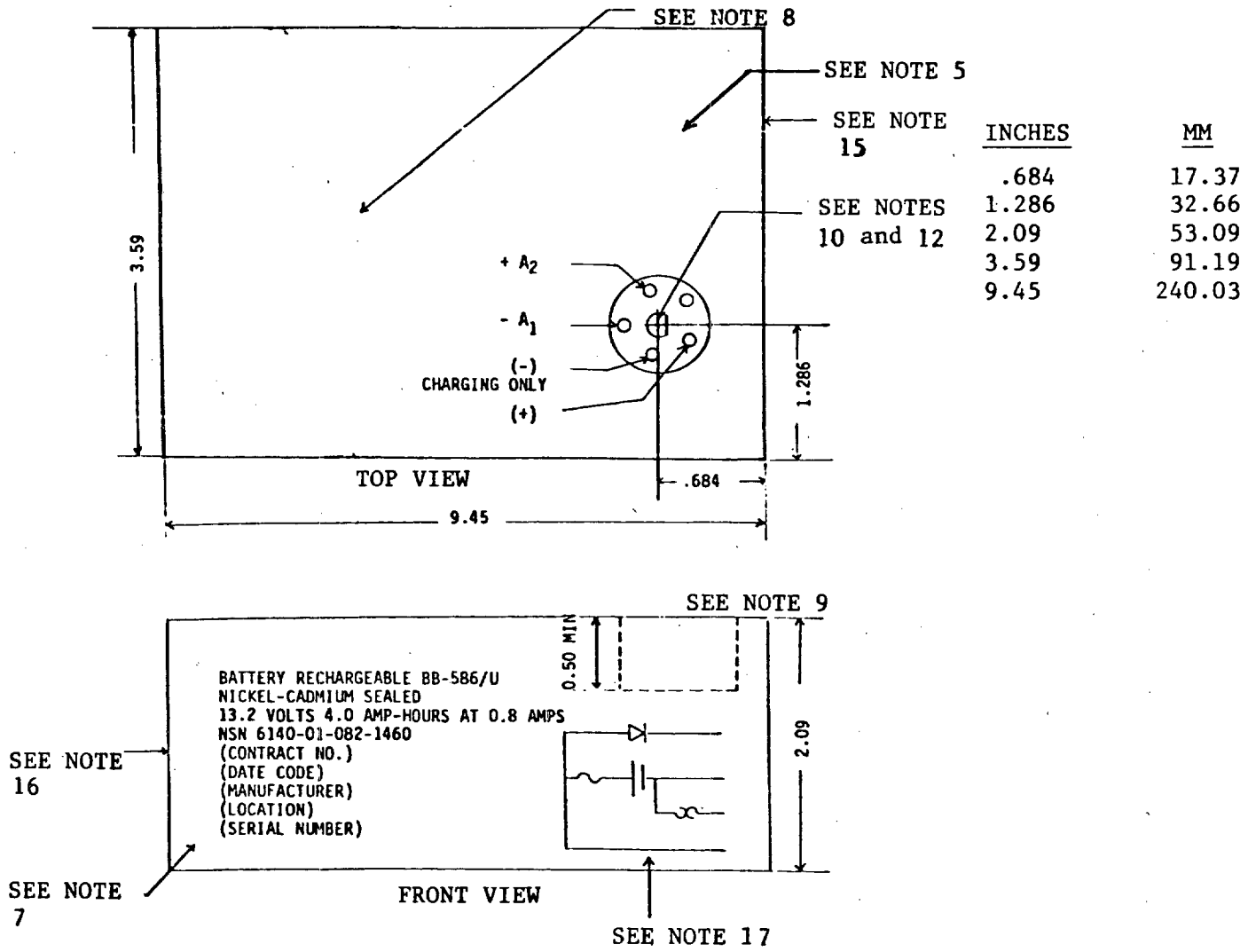
MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED NICKEL CADMIUM BB-586/U

This specification sheet is approved for use by the Electronic Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sealed nickel cadmium rechargeable battery type described herein shall consist of this document and the latest issue in effect of Specification MIL-B-49436(ER).

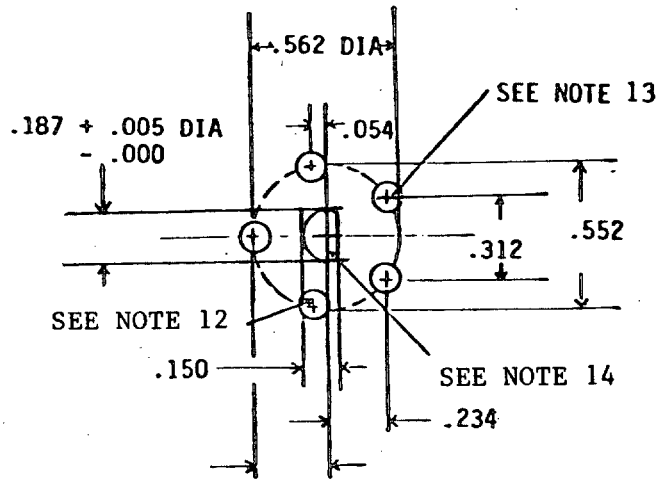


INCHES	MM
.684	17.37
1.286	32.66
2.09	53.09
3.59	91.19
9.45	240.03

FIGURE 1. BB-586/U BATTERY

FSC-6140

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INCHES	MM
.005	.13
.054	1.37
.150	3.81
.187	4.75
.234	5.94
.312	7.92
.552	14.02
.562	14.27

FIGURE 2. TERMINAL DIMENSIONS

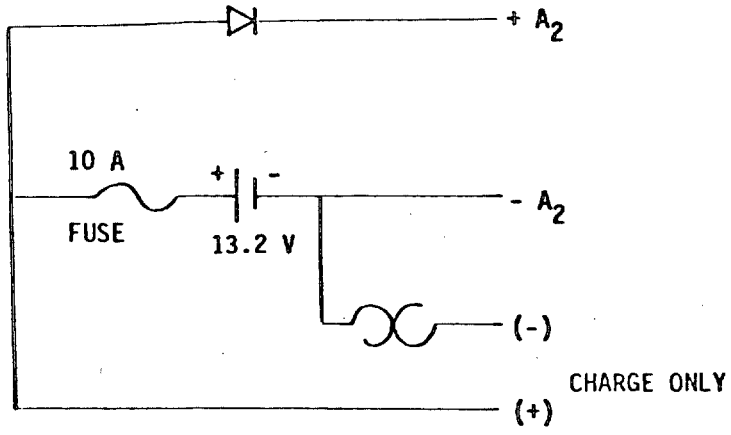


FIGURE 3. WIRING SCHEMATIC

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DRAWING NOTES

1. All dimensions in inches.
2. Tolerance on two place (.00) decimals: $\pm \begin{matrix} .03 \\ .06 \end{matrix}$ unless otherwise specified.
3. Tolerance on three place (.000) decimals: ± .005 unless otherwise specified.
4. Dimensioning shall be per USAS-114.5 Y14.5 1966.
5. Last date charged label shall be located on top face of battery.
6. A pressure relief machanism shall be located in the case, with the preferred location under the connector. It shall release gas before case damage is incurred.
7. Stencil (or equivalent) identification as shown. Manufacturer must fill in applicable information in parenthesis. Marking color shall be white.
8. Stencil (or equivalent) charging instructions in the location shown. Color marking white. Text of instructions must be submitted for approval.
9. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the case, nor be more than 1/16 inch below the adjacent outside surface of the case both before and after all testing.
10. Pin-circle center.
11. Socket must float 3/32 ± 1/32 in all directions from the specified pin center location in the planes of the socket.
12. Five holes to fit 1/8 inch pins, 7/16 inch in length on male plug, per drawing SM-C-447075.
13. Socket insert contact not required in this hole.
14. "D" hole for plug positioning.
15. Fuse location, this face of battery. Fuse holder shall be flush with battery case.
16. Spare fuse location this face of battery. Fuse holder shall be flush with battery case. Stencil (or equivalent) legend "spare fuse." Marking color shall be white.
17. Stencil (or equivalent) wiring schematic (see Figure 1) in location shown. Color marking shall be white.

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REQUIREMENTS:

Rated capacity (C_5):	4.0 AH
Voltage (Nominal):	13.2 Volts
Weight (Maximum):	1.8 Kg (4 lbs)
Connector:	Floating socket type (see Figure 2)
Number and type of cells:	11 sealed "D" size or equal in series
Charge rate for testing and on battery (C/4 for 6 hrs):	1.0 amps

Discharge Rates	5 Second Voltage(V)	Final Voltage(V)	Current (Amps)
C/5	-	11.0	0.8
C_5	-	10.0	4.0
2.5 C_5	12.1	10.0	10.0

Test	Discharge Rate	Capacity Requirement(AH)
Full Capacity Discharge	C_5	3.5
Cycle Life, 224 cycles	C_5	2.8
Overcharge, 24 hours	C_5	4.0
High Rate Discharge	2.5 C_5	2.8
Low Temperature Discharge at -18°C (0°F)	C_5	2.0
Retention of Charge 7 days at 51°C (124°F)	C_5	2.0
Pulse Discharge	Not applicable	-
Vibration	C/5	4.0

NOTES - 1. Unless otherwise specified, all values are minimum.
2. All charges and discharges shall be performed on fully assembled batteries through the terminals.

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REQUIREMENT:

Mounting screws for connectors. Not applicable.

Over temperature protection. Batteries shall contain a thermal cutout device designed to prevent charging at temperatures above 49°C (120°F). Charging shall not be inhibited at temperatures between 5 and 38°C (40 and 100°F).

Reversal protection. Batteries shall contain a diode in the discharge line to prevent charging through the discharge circuitry. The diode shall be capable of conducting up to 10A continuously without failure or excessive voltage drops.

Overload protection. The battery shall be protected from physical and electrical damage from any overload conditions (including a direct short circuit), in either the charge or discharge connections by the use of fuses or circuit breakers. Fuses or breakers shall carry a minimum of 10 amps for 17 minutes without opening. If fuses are used, at least one spare shall be provided.

Electrical leakage. No potential shall be obtained between terminals +A2 and -A2, and between the + and - charge terminals. Voltage readings shall be taken between terminals +A2 and -A2, and the + and - charge terminals after the fuses or circuit breaker is removed.

Connector. The battery connector shall be a floating type which shall be capable of withstanding a minimum of 500 coupling and uncoupling with charging cable with no breakage, distortion or loss of electrical performance.

TESTS:

Overcharge test. The battery shall be charged per 4.6.1 at 1.0 amperes for 24 hours. The battery voltage and temperature shall be recorded. The thermal cutout device shall meet the over temperature protection requirements specified. The battery shall meet the Overcharge Test requirements specified on the applicable specification sheet.

High rate discharge test. The battery shall be charged per 4.6.1 and discharged at 10 amperes and meet the requirements specified for the High Rate Discharge Test. The fuse or circuit breaker and diode shall meet the requirements specified for the overload protection and the reversal protection respectively.

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TESTS (CONTINUED):

Reversal protection test. Apply a 32 ±1 volt charging source to the +A2 and -A2 terminals for a period of 1 minute. A milliammeter in the circuit shall be read at the end of one minute to determine that no current in excess of 0.1 milliamperes flows.

CUSTODIAN:
ARMY-ER
REVIEWER:
ARMY- CR, AV, GL

PREPARING ACTIVITY:
ARMY-ER
(PROJECT 6140-A577-3)

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